Gradient-Geometry Baselines ($f = 0$) and Their Correlation with Robustness Under Attack

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Contents

1 Setup

1.1 Gradient-geometry baselines

Three models are trained on MNIST with $n = 10$ honest clients, $f = 0$ (no Byzantine clients, `NoAttack`), aggregator `TrMean` with pre-aggregators `NNM+ARC`, across four data-heterogeneity levels $\gamma \in \{1.0, 0.66, 0.33, 0.0\}$ (`gamma_similarity_niid`), for 500 training steps, 1 seed:

- **SNN (atan, $\alpha = 1.2$)**: `cnn_mnist_snn`, LIF layers, surrogate gradient `atan`, $\beta = 0.95$, threshold $= 1.0$, $T = 10$ time steps, lr $= 0.10$.
- **CNN ReLU**: `cnn_mnist`, lr $= 0.15$.
- **CNN Tanh**: `cnn_mnist_tanh`, lr $= 0.15$.

All three models have $d = 431{,}080$ parameters (identical architecture, only the activation/neuron model differs). Metrics are computed online, every training step, directly on the honest, post-momentum gradient vectors actually used for aggregation — no raw vectors are ever logged to disk.

1.2 Robustness sources (existing sweeps)

These robustness sweeps were originally run on a separate Docker host; only a *subset* of the full grid was imported into this local repository, not the complete sweep output. As a result, no locally-available sweep matches all of the geometry baseline's hyperparameters (aggregator, learning rate, SNN α) with complete $f \in \{0, \ldots, 5\}$, γ, and attack coverage — only whatever combination of (aggregator, lr/α, attack) happened to be imported is usable here. The best available *common* aggregator with full $(f, \gamma, \text{attack})$ coverage for all three models among the imported data is `GeometricMedian+NNM+ARC` (not `TrMean`, which is only fully covered for CNN Tanh). The sources actually used:

Model	Results directory	Deviation from geometry baseline
SNN atan	`results/snn/robust_new_atan_sweep`	$\alpha = 1.25$ (only $\alpha \in \{0.5, 0.75, 1.0, 1.25, 1.5, 2.0, 3.0\}$ were imported, not 1.2; 1.25 is the closest); only `Optimal_ALittleIsEnough_neg1` was imported for this combo (single-attack, not worst-of-3)
CNN ReLU	`results/cnn/weekend`	lr$= 0.05$ (the lr$= 0.15$ import, `robust_comparison_sweep`, only covers $f = 0..2$); worst-of-3 attacks
CNN Tanh	`results/cnn/tanh_heatmap_sweep`	lr$= 0.15$, exact match; worst-of-3 attacks

For each (f, γ) cell, the accuracy score is the mean test accuracy over the last three evaluation checkpoints (steps 400, 450, 500), averaged over 5 training seeds, then the *worst case over available attacks* (minimum), matching the convention already used by the repository's own `test_heatmap` utility. This substitution of aggregator/lr/α means the robustness numbers below are a *proxy*, not an exact match to the geometry-baseline training dynamics; the geometry metrics themselves come only from the exact-match $f = 0$/`TrMean` runs described above.

2 Metric Definitions

Let $\{m^{(k)}\}_{k=1}^{n}$ be the honest, post-momentum gradient vectors at a given training step ($n = 10$). For each coordinate j:

$$\mu_j = \frac{1}{n}\sum_{k=1}^{n} m_j^{(k)} \qquad \sigma_j = \sqrt{\frac{1}{n}\sum_{k=1}^{n}\left(m_j^{(k)} - \mu_j\right)^2} \tag{1}$$

$$\mathrm{CV}_j = \frac{\sigma_j}{|\mu_j| + 10^{-12}} \qquad A_j = \frac{\max\bigl(\#\{k : m_j^{(k)} > 0\},\ \#\{k : m_j^{(k)} < 0\}\bigr)}{n} \tag{2}$$

where $|m_j^{(k)}| < 10^{-12}$ abstains from the sign count in A_j (denominator n stays fixed).

Scalar summaries (one value per training step):

$$N = \frac{\mathrm{std}_k\left(\|m^{(k)}\|_2\right)}{\mathrm{mean}_k\left(\|m^{(k)}\|_2\right) + 10^{-12}} \qquad \text{(dispersion of per-client gradient norms)} \tag{3}$$

$$Q = \frac{1}{d}\sum_{j=1}^{d} \mathbb{1}[A_j \geq 0.8] \qquad \text{(raw fraction of coordinates in sign-consensus)} \tag{4}$$

$$S = \frac{\sum_j \mu_j^2 \cdot \mathbb{1}[A_j \geq 0.8]}{\sum_j \mu_j^2 + 10^{-12}} \qquad \text{(signal-weighted consensus)} \tag{5}$$

$$\cos_{\mathrm{mean}} = \underset{k<l}{\mathrm{mean}} \frac{\langle m^{(k)}, m^{(l)} \rangle}{\|m^{(k)}\|\|m^{(l)}\|} \qquad \text{(mean pairwise cosine similarity)} \tag{6}$$

Per-layer breakdown: for a layer L spanning coordinate range $[a_L, b_L)$, Q_L and S_L are computed identically but restricted to that range. S_{layer0} denotes the first weight-bearing layer (`conv1` for the SNN, `_c1` for the CNNs); $S_{\mathrm{deep\_mean}}$ is the mean of S_L over all other layers.

Summary statistic used below: for each (model, γ) run, we take the *median* of each per-step metric over steps $\in [100, 400]$ (the "established regime" window, chosen after visually inspecting the trajectories – see Section 3).

3 Geometry Trajectories

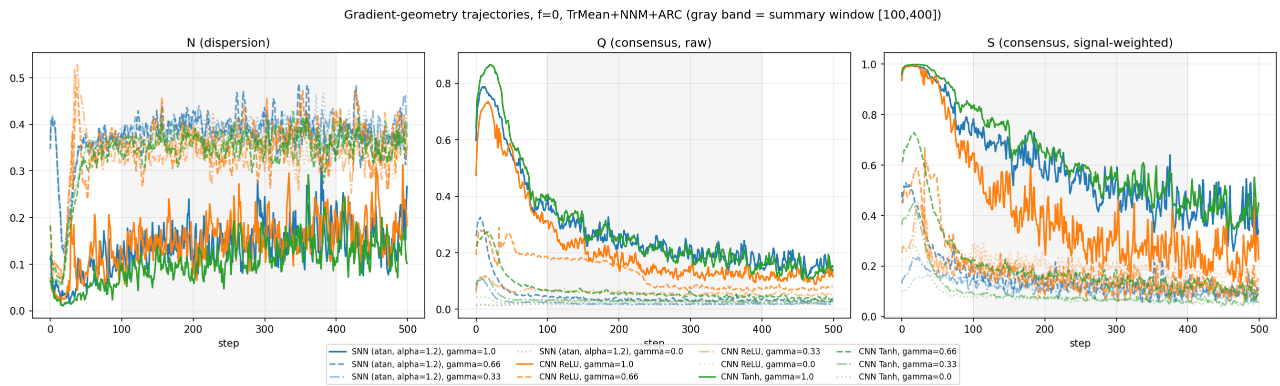

Figure 1: $N(\tau)$, $Q(\tau)$, $S(\tau)$ over training, one curve per (model, γ). Solid = $\gamma = 1.0$, dashed = 0.66, dash-dot = 0.33, dotted = 0.0. Blue = SNN, orange = CNN ReLU, green = CNN Tanh. Gray band marks the [100,400] summary window.

Observed fact: at $\gamma = 1.0$ (solid curves), S keeps decreasing slowly through the entire [100,400] window rather than reaching a stable plateau; at $\gamma < 1.0$, all three models reach a noisy plateau by step $\sim$150. The [100,400] median is therefore a "passing value" rather than a true steady state for the $\gamma = 1.0$ runs.

3.1 Trajectories split by γ (direct 3-model comparison)

The figure above overlays all 12 (model, γ) curves per panel, which makes it hard to compare the 3 models at a fixed γ. Figures 2–5 below show the same $N(\tau)$, $Q(\tau)$, $S(\tau)$ data, one figure per γ, with only the 3 models overlaid (blue = SNN, orange = CNN ReLU, green = CNN Tanh).

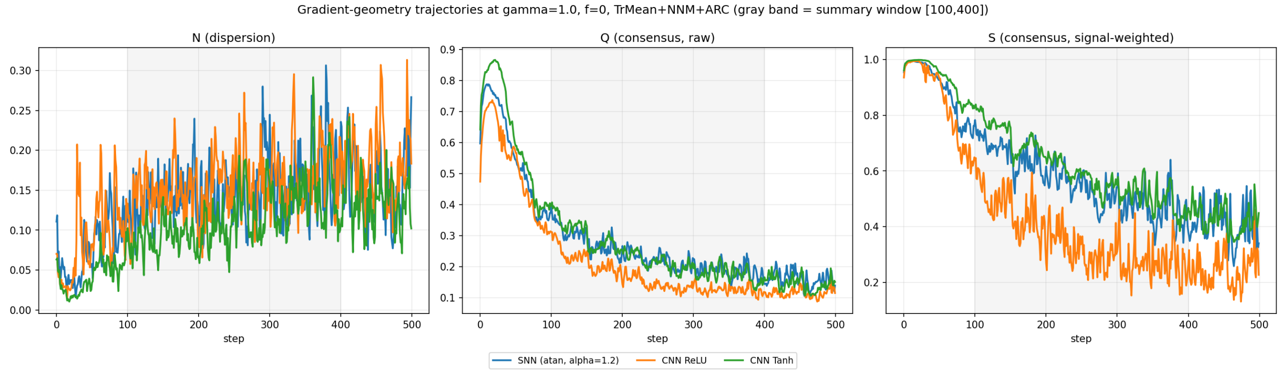

Figure 2: $\gamma = 1.0$: the 3 models compared directly.

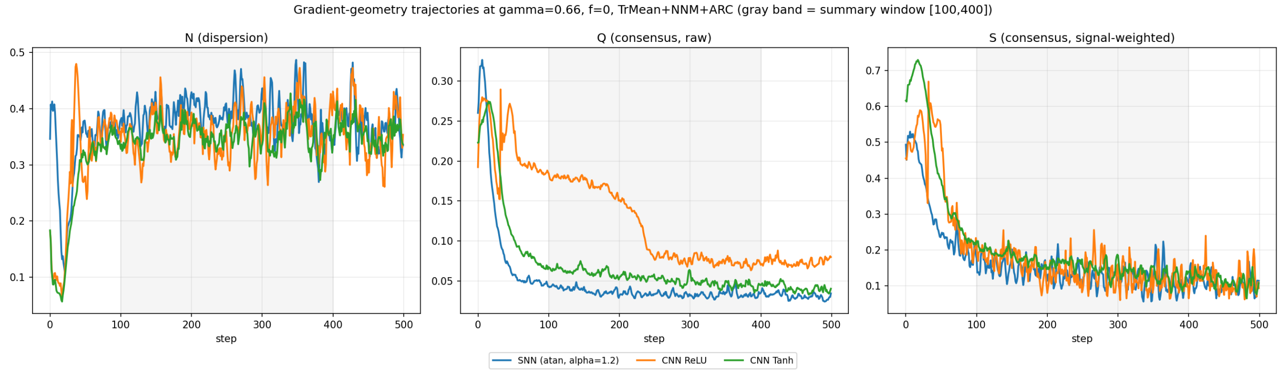

Figure 3: $\gamma = 0.66$: the 3 models compared directly.

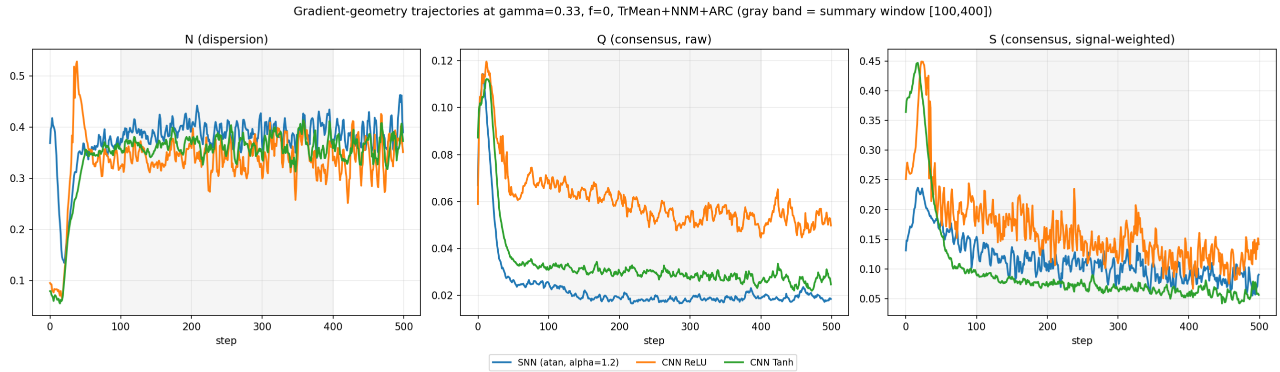

Figure 4: $\gamma = 0.33$: the 3 models compared directly.

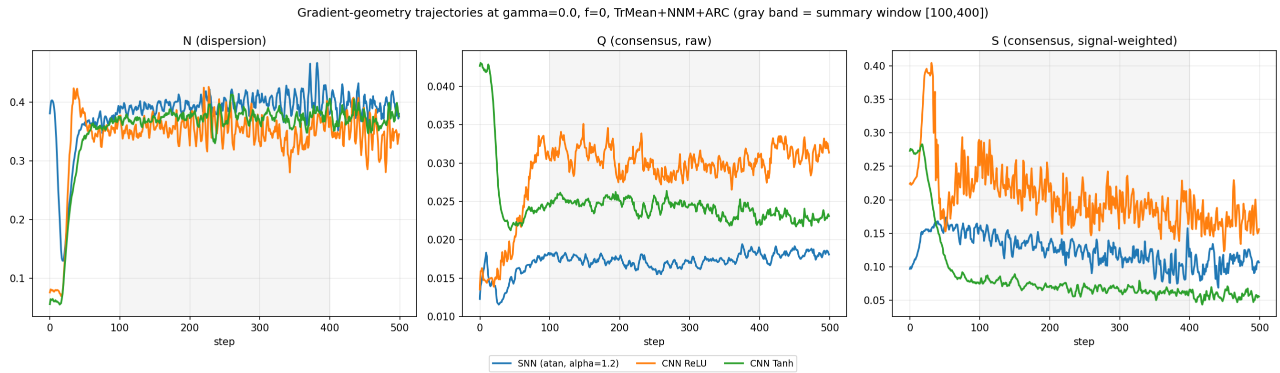

Figure 5: $\gamma = 0.0$: the 3 models compared directly.

4 Geometry Summary Table

Model	γ	N	Q	S	S_{layer0}	$\cos_{\text{mean}}$
SNN atan	1.0	0.143	0.223	0.547	0.624	0.085
SNN atan	0.66	0.388	0.034	0.132	0.161	−0.074
SNN atan	0.33	0.391	0.019	0.108	0.055	−0.086
SNN atan	0.0	0.395	0.017	0.124	0.048	−0.087
CNN ReLU	1.0	0.158	0.143	0.331	0.648	0.004
CNN ReLU	0.66	0.362	0.085	0.137	0.069	−0.082
CNN ReLU	0.33	0.341	0.057	0.150	0.018	−0.093
CNN ReLU	0.0	0.354	0.030	0.198	0.012	−0.093
CNN Tanh	1.0	0.106	0.216	0.591	0.552	0.107
CNN Tanh	0.66	0.352	0.053	0.155	0.145	−0.071
CNN Tanh	0.33	0.365	0.029	0.072	0.025	−0.090
CNN Tanh	0.0	0.374	0.025	0.068	0.014	−0.092

Median over steps $\in [100, 400]$. Full table with q25/q75 in `summary_table.csv`.

4.1 Registered predictions, checked against these numbers

Prediction	Verdict
$S_{\text{SNN}} < S_{\text{ReLU}}$ and $S_{\text{SNN}} < S_{\text{Tanh}}$ at every γ	Holds only at $\gamma = 0.66$. At $\gamma = 1.0$, $S_{\text{ReLU}} = 0.331$ is the *lowest* of the three, not the SNN (0.547).
$N_{\text{ReLU}} \gg N_{\text{SNN}} \approx N_{\text{Tanh}}$	Violated at 3 of 4 γ values; only $\gamma = 1.0$ confirms it. Elsewhere all three models have near-identical N (see Section 3: N clusters by γ, not by model).
S decreases monotonically as γ decreases, for all models	Holds only for CNN Tanh. SNN and ReLU both tick up slightly between $\gamma = 0.33$ and $\gamma = 0.0$.
$S_{\text{layer0}}(\text{SNN}) \gg S_{\text{deep\_mean}}(\text{SNN})$	Holds only at $\gamma = 1.0$ (0.624 vs. 0.579, small margin). At $\gamma < 1.0$ the deep layers have *higher* consensus than layer0, with the gap widening as $\gamma \to 0$.

5 Robustness Heatmaps

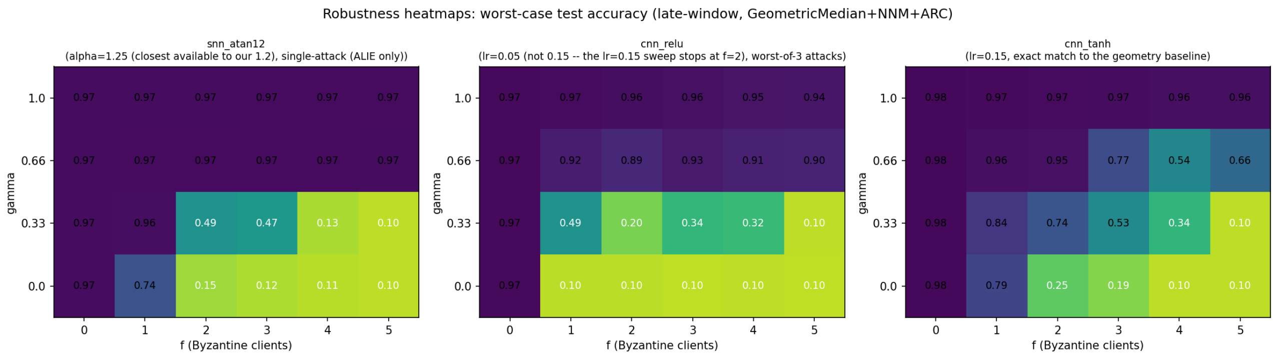

Figure 6: Worst-case (min-over-attacks) test accuracy, late-window average, per (model, f, γ). Aggregator `GeometricMedian+NNM+ARC`. See Section 1.2 for the exact hyperparameter deviations per model.

Observed facts: all three models are near-perfectly robust at $\gamma = 1.0$ across every f tested (accuracy stays ≥ 0.94). At $\gamma \leq 0.66$, accuracy collapses toward ~0.10 (chance level on 10 classes) once $f \geq 2$–3, for all three models. CNN Tanh degrades more gradually at $\gamma = 0.66$ (e.g. $0.96 \to 0.54$ across $f = 0..4$) than SNN or CNN ReLU, which drop sharply.

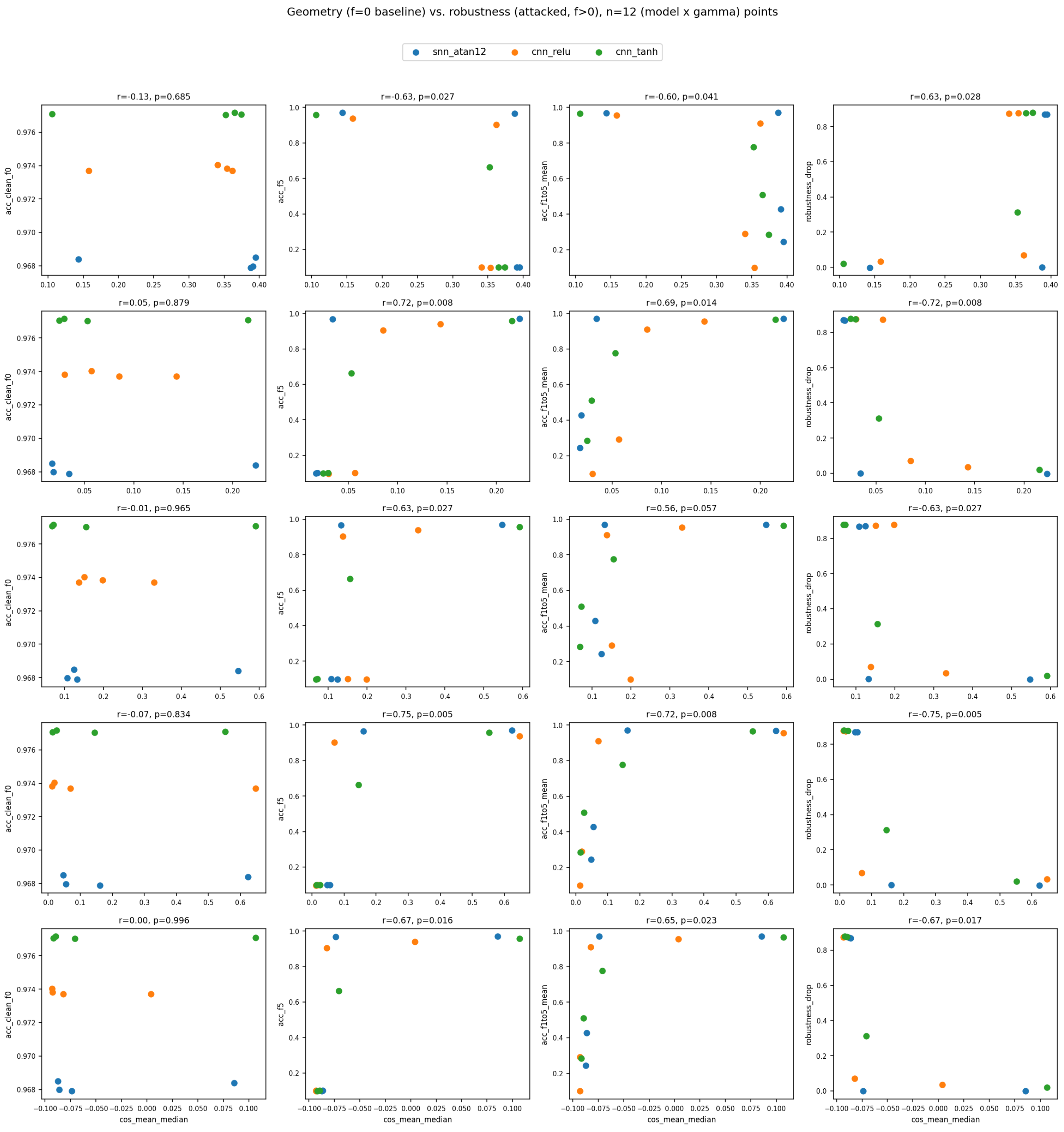

Figure 7: Each panel: one geometry metric (median, $f = 0$, steps [100,400]) vs. one robustness summary, $n = 12$ points (3 models $\times$ 4 γ). r = Pearson correlation, p = two-sided p-value.

6 Correlation: Geometry ($f = 0$) vs. Robustness (attacked, $f > 0$)

Geometry metric	Robustness metric	r	p	n
S_{layer0}	acc_f5	0.751	0.005	12
S_{layer0}	robustness_drop	−0.750	0.005	12
Q	acc_f5	0.725	0.008	12
S_{layer0}	acc_f1to5_mean	0.724	0.008	12
Q	robustness_drop	−0.724	0.008	12
$\cos_{\text{mean}}$	acc_f5	0.673	0.016	12
N	acc_f5	−0.633	0.027	12
S	acc_f5	0.633	0.027	12
N	acc_clean_f0	−0.131	0.685	12
S_{layer0}	acc_clean_f0	−0.068	0.834	12
Q	acc_clean_f0	0.049	0.879	12
S	acc_clean_f0	−0.014	0.965	12
$\cos_{\text{mean}}$	acc_clean_f0	0.001	0.996	12

Full table (20 metric pairs) in `correlation_table.csv`. `acc_f5` = worst-case accuracy at $f = 5$; `acc_f1to5_mean` = mean worst-case accuracy over $f = 1..5$; `robustness_drop` = `acc_clean_f0` − `acc_f5`.

Observed fact: every geometry metric computed at $f = 0$ correlates moderately-to-strongly ($|r| = 0.63$ to 0.75, $p < 0.03$) with robustness metrics measured under attack ($f > 0$) at the *same* γ, across a 12-point sample (3 models × 4 γ). None of these same metrics correlates with clean, attack-free accuracy at $f = 0$ ($|r| < 0.14$, $p > 0.68$ in all five cases).

7 Interpretation

The strongest, most consistent signal in this data is not a difference between architectures (SNN vs. ReLU vs. Tanh) but an almost universal effect of γ (data heterogeneity) across all three: it drives N, $\cos_{\text{mean}}$'s sign flip, and the per-layer consensus redistribution seen in Section 4.1. The four originally registered predictions, which were architecture-centric, mostly failed outside the iid case ($\gamma = 1.0$) – so architecture differences in gradient geometry seem to be a $\gamma = 1.0$-only phenomenon, not a general property of these models.

The correlation results are consistent with a specific, testable story: *how much the honest clients agree with each other at $f = 0$ (in sign, and directionally via cosine similarity) predicts how much slack the aggregator has left over to absorb Byzantine attackers, without predicting how well the model learns in the first place.* Concretely, S and Q (both consensus proxies) and $\cos_{\text{mean}}$ correlate positively with post-attack accuracy, while N (dispersion) correlates negatively – exactly the sign pattern one would expect if geometric agreement at $f = 0$ is a leading indicator of the margin an aggregator like `TrMean`/`GeometricMedian` has before Byzantine votes can dominate a coordinate. The complete absence of correlation with clean accuracy ($f = 0$, no attack) suggests these metrics are specifically informative about *robustness headroom*, not general model quality – which is the more useful (and non-trivial) property for this research question.

Two caveats limit how far this should be pushed. First, $n = 12$ with 3 models × 4 γ means these 12 points are not independent draws – much of the correlation could be a γ-driven confound shared by both geometry and robustness, rather than geometry being causally informative beyond γ alone; a partial-correlation or within-model check would be the natural next step. Second, the robustness numbers are a proxy (Section 1.2): different aggregator (`GeometricMedian` instead of `TrMean`), different SNN α (1.25 vs. 1.2), different CNN ReLU learning rate (0.05 vs. 0.15), and a single attack for the SNN row. The direction and rough size of the correlations are unlikely to be pure artifacts of these substitutions, but an exact-match sweep (same `TrMean`+`NNM`+`ARC`, same lr/α, all 3 attacks, $f = 0..5$) would be needed to confirm the effect sizes precisely.